DS	**Session**	Linked List	**Question Information**	Level 1 Challenge 32

Question description

saran, subash, and Yasir alias Pari are three first-year engineering students of the State Technical Institution (STI), India. While saran and subash are average students who come from a Middle class, Yasir is from a rich family. saran studies, engineering as per his father's wishes, while subash, whose family is poor, studies engineering to improve his family's financial situation.

Yasir, however, studies engineering of his simple passion for developing android applications.

Yasir is participating in a hackathon for android application development. the task is Insertion in a Doubly Linked list at beginig.

Functional Description:

In the doubly linked list, we would use the following steps to insert a new node at the beginning of the doubly linked list.

- Create a new node
- Assign its data value
- Assign newly created node’s next ptr to current head reference. So, it points to the previous start node of the linked list address
- Change the head reference to the new node’s address.
- Change the next node’s previous pointer to new node’s address (head reference)

Constraints

0<N<100

0<arr<1000

Input Format

First line indicates the number of elements N to be inserted in array

Second line indicates the array elements according to the N

Output Format

First line represents the doubly linked list in forward direction

Second Line represents the doubly linked list in backward direction

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
16
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
```

EXPECTED OUTPUT

```
17 98 32 56 24 23 19 18 14 13 11 9 7 4 3 1
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
```

Test Case 2

INPUT (STDIN)

```
15
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
```

EXPECTED OUTPUT

```
2 1 0 2 2 0 2 0 1 1 1 0 2 1 0
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
struct Node
```

Test Case 2

KEYWORD

```
struct Node *next;
```

Test Case 3

KEYWORD

```
struct Node *prev;
```

Test Case 4

KEYWORD

```
void insertStart(struct
Node** head,int data)
```

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
2	269	47